

Domain Adaptation for Betting With Betfair Signals and TAB Execution

Executive summary

A system that trains on Betfair exchange data, queries Betfair live, but executes bets on TAB without programmatic TAB market data creates a **partial-observability domain shift**: you observe a rich, high-frequency exchange state (order book, traded volume, spreads) yet must make decisions for a bookmaker/tote environment where the actionable variable is the **price TAB will actually accept** (and that price may be fixed-odds, tote, or a hybrid product depending on channel and jurisdiction). Betfair provides a low-latency streaming feed and rich market microstructure fields (best offers, full ladders, traded ladders, last-traded price, market definitions, and price projections). ¹ TAB's own help and rules emphasise that tote dividends are **approximate and can change**, with TAB taking a stated commission from pools; and that fixed odds are "fixed at the time the bet is placed" but still subject to rule-based adjustments (e.g., scratchings/dead heats) and discretionary acceptance (refusal, limits, price changes). ²

The most practical architecture separates the problem into two coupled models:

1) **Outcome model**: estimate the underlying win probability (or distribution over outcomes) using Betfair live features plus stable, domain-invariant covariates (race metadata, runner fundamentals). This part can be trained and evaluated using publicly observable labels (race results) and does not require TAB odds. Domain adaptation here mainly concerns **time non-stationarity** and changes in the relationship between market state and outcomes. ³

2) **Price translation model**: estimate the distribution of **TAB executable odds** (accepted fixed odds and/or final tote dividend) conditioned on Betfair state and context. Without a TAB feed, this must be learned from **human-collected TAB snapshots** (manual entry, controlled capture) or via **licensed data access** (official API or vendor). TAB's web services exist, but their personal-use terms restrict reuse, storage, derivative works, and third-party access, and they explicitly prohibit screen scraping on account channels; commercial use requires additional approvals. ⁴

Decisions become robust when you treat TAB odds as a **random variable** and bet only when expected value is positive under conservative assumptions (e.g., lower confidence bound on TAB odds). Conformal methods under covariate shift provide a principled way to build such conservative bounds when you have unlabeled target covariates (here: current Betfair states) and can estimate density ratios. ⁵

Legally, the highest-confidence path is to pursue **official/contractual access** (TAB Studio commercial pathway and/or licensed racing data vendors) rather than scraping. The regulatory environment for online wagering in Australia includes explicit federal restrictions (Interactive Gambling Act) enforced by the Australian Communications and Media Authority ⁶; online wagering is legal when providers are appropriately licensed, but some online betting forms are prohibited. ⁷

Betfair-TAB domain gap

Betfair and TAB differ along dimensions that matter directly for transfer learning, feature mapping, and operational risk.

Market structure and price formation

Betfair's exchange model supports **back and lay** and relies on matching orders from other market participants; your effective price is determined by available offers and matching. ⁸ TAB tote betting is explicitly pool-based: customers bet into a pool, TAB removes a stated commission (varying by bet type), and dividends depend on how the pool is distributed among winners. ⁹ TAB fixed odds is explicitly described as bookmaking: a quoted price is "fixed at the time the bet is placed" (subject to scratchings/dead heats), while rules also allow price changes and discretion in acceptance. ¹⁰

Liquidity and observable microstructure

Betfair exposes supply/demand and liquidity via ladders and traded volumes (best offers, full ladders, traded ladders, LTP, market definitions; with ladder depth controlled by request parameters). ¹¹ TAB tote exposes pool information (e.g., prize pools) alongside approximate dividends in the website/venue information systems. ¹² TAB fixed odds generally does not expose an order book; instead, the bookmaker can refuse bets and impose stake/payout limits, which creates "liquidity" that is discretionary and partially hidden. ¹³

Odds formats, tick/price constraints, and margin

Betfair enforces discrete "acceptable" odds increments across ranges (a tick ladder), constraining both modelling and any price-based feature engineering (e.g., tick distance, spread in ticks). ¹⁴ Betfair charges commission on **net winnings** on a market, calculated by multiplying net winnings by the market base rate. ¹⁵ TAB tote returns depend on pool commission ("fixed commission out of the pool" with a stated range by bet type). ⁹ TAB fixed odds embeds bookmaker margin via overround rather than post-hoc commission, and (empirically) includes pricing constraints such as rounding and discretionary movement; the platform publishes fixed-odds fluctuation history as a consumer feature, signalling that fixed odds prices move pre-event. ¹⁶

Latency, live operation, and bet acceptance

Betfair's stream API is positioned as low-latency market-data access. ¹⁷ Betfair in-play introduces a platform delay (stated as 1–12 seconds) for in-play markets. ¹⁸ Betfair also notes that "virtual bets" (used to construct a fuller display) may update later than non-virtual prices (on the order of ~150 ms). ¹⁹ TAB explicitly frames currency-of-display as best endeavours rather than a guarantee (and limits liability for errors/omissions in market display). ¹³ For a manual/tab-machine workflow, additional human latency creates systematic slippage between the Betfair snapshot you used for inference and the TAB price actually accepted.

Market rules: non-runners, deductions, cancellations

Non-runner handling differs and must be explicitly modelled.

- Betfair exchange uses reduction factors; rules specify when reductions apply and when unmatched lay offers can be cancelled depending on reduction factor thresholds and market type. ²⁰
- TAB fixed odds uses a published deductions schedule for NSW/Victoria fixed odds with “cents in the dollar” deductions by price ranges (separate win/place deduction columns). ²¹
- TAB conditions emphasise rights to refuse bets and limitations on cancellation once accepted. ²²

Visual intuition

Prioritised actionable options with pros, cons, and tradeoffs

The constraint “TAB data access is currently unavailable” forces a choice between (a) making TAB data available legally, or (b) explicitly treating TAB odds as latent and collecting minimal labels via humans. TAB’s own documentation indicates that **TAB Studio Web Services API** can provide betting information and betting operations across multiple jurisdictions, but access and permitted use depend on account tier and terms; commercial use is treated separately. ²³

Option set overview

Option	What you do	Practical upside	Core risks / downsides	Legal posture
Official TAB data access (TAB Studio commercial)	Apply as “commercial user” and obtain approvals; integrate API to pull odds & pools	Highest fidelity TAB features; automation; easier evaluation	Approval lead time; fees; compliance obligations; still channel/ jurisdiction complexity	Strongest, if contracted and compliant ²⁴
Licensed data vendors / partnerships	Buy/feed odds (TAB fixed, tote pools) and race info via vendor ecosystem	Avoid building your own collection; scalable	Ongoing cost; vendor coverage gaps; licensing constraints on redistribution	Typically strong if licensed; still contractual constraints
Manual capture + active learning	Build logging UX; humans enter accepted TAB odds (and optionally pre-bet displayed odds) for selected events	Cheapest to start; works under strict “no API” constraints	Labour; sparse labels; bias (you only log what you consider betting); slower iteration	Strong if you only record your own observed quotes/ tickets; still avoid ToS violations

Option	What you do	Practical upside	Core risks / downsides	Legal posture
Retail-terminal OCR / screen capture	Capture terminal screen and OCR odds	Potentially increases label volume	OCR error; operational burden; strong ToS risk if it constitutes prohibited scraping/replication	High risk because of explicit prohibitions on screen scraping and derivative use in TAB materials ²⁵
Pure approximation (no TAB labels)	Bet using Betfair prices as proxy for TAB	No data collection	High uncertainty; fragile; hard to validate; may overbet	Legally low risk, technically high risk

Recommended priority order under unspecified budget

A defensible build sequence is:

First: implement the two-model architecture and begin **manual capture + active learning** immediately (so you start collecting ground truth for translation). Active learning literature supports prioritising labels where the model is uncertain or where utility is high. ²⁶

Second: pursue **official commercial access** in parallel as a medium-term capability, because it unlocks automated pricing features and lowers long-run operational risk. TAB explicitly separates “account holder personal use” and “commercial users” registration for API usage. ²⁴

Third: if commercial API access is slow or incomplete, consider **licensed vendor feeds** for odds and race data; TAB’s own terms identify multiple third-party copyright owners for racing materials and signal that approvals and licensing are normal in this ecosystem. ²⁷

Data blueprint and feature mapping

Core modelling objects

A working schema needs to represent:

- event identity (track/venue, race number, scheduled start)
- runners/selections
- Betfair market states over time (snapshots and/or stream deltas)
- TAB observed odds/pools (manual or licensed feed)
- your decision and execution (stake, bet type, time, accepted odds)
- outcomes (winner/place, payouts)

Betfair's API surfaces market-level and runner-level traded volume, full ladders, LTP, and SP projections, suitable for time-series features. ²⁸ TAB tote and approximate-dividend pages strongly imply that approximate dividend systems also track prize pools and a range of bet types. ¹²

Required schema (example tables)

Table	Key fields	Notes
event	event_id, jurisdiction, meeting_name, start_time_utc, type	Include jurisdiction because TAB rules/offers vary by state/territory. ²⁹
runner	event_id, runner_id, runner_name, barrier, jockey, trainer	Keep a stable runner_id separate from Betfair selectionId and TAB runner number.
betfair_market	event_id, market_id, market_type, timezone, status	Market definitions are available via Betfair's market-def updates. ¹⁹
betfair_snapshot	market_id, selection_id, t, atb_ladder[], atl_ladder[], traded_vol[], ltp, tv_total, spread_ticks, imbalance	Ladders and traded ladders are first-class data in listMarketBook/stream. ¹⁹
tab_quote	event_id, t, channel (retail/app/web), product (fixed/tote), runner_number, display_odds, pool_win, pool_place, notes	Approximate dividend + prize pool availability is described by TAB. ¹²
execution	event_id, t_exec, product, runner_number, stake, accepted_odds, ticket_id, cancelled_flag	Include whether TAB refused/repriced; TAB rules allow refusal and price changes. ³⁰
outcome	event_id, runner_id, finish_pos, payout_fixed, payout_tote	Payout depends on product and rules; tote depends on pools; fixed depends on odds & deductions. ³¹

Mapping Betfair → TAB features (example)

The goal is to build a translation layer that approximates TAB price and acceptability constraints from observable Betfair dynamics.

Betfair observable	Why it helps	Approximate TAB analogue
Best back/lay prices and spread in ticks	Encodes current consensus, uncertainty, and liquidity; tick ladder is discrete and stable. ³²	Expected fixed-odds quote level; probability of price movement before acceptance
Order book depth / ladder shape	Captures supply/demand imbalance and resilience. ¹⁹	Risk of TAB shortening/lengthening; surrogate for bookmaker risk appetite signals
Traded volume trajectory near jump	Proxy for information arrival and crowd conviction. ³³	Likely tote pool concentration changes; likelihood of late tote “crunch”
SP projected / near-far prices	Betfair explicitly exposes SP projections; SP concept is positioned as similar to tote. ³⁴	Expected tote dividend distribution at jump; mapping anchor between exchange and pool pricing
Market-definition events (suspension/in-play)	Indicates time boundaries and event state. ³⁵	TAB cut-off times; risk of price freeze / bet rejection close to start

Converting prices to comparable probability space

A practical mapping step is to convert all prices to **implied probabilities**, removing platform-specific fees structurally:

Betfair back bet, commission-adjusted break-even probability

If odds are O (decimal) and commission on net winnings is c , a break-even probability is:

$$p_{\text{BE}} = \frac{1}{1 + (O - 1)(1 - c)} = \frac{1}{(1 - c)O + c}$$

Commission on net winnings and market base rates are explicit in Betfair’s charges documentation. ¹⁵

TAB tote: published description implies: pool $\rightarrow$ take-out/commission $\rightarrow$ remaining pool divided among winners, so implied probabilities from displayed approximate dividends depend on pool shares and take-out; the system also publishes prize pools as part of its approximate dividend info. ³⁶

TAB fixed odds: bookmaker margin is embedded in quoted odds; scratchings trigger structured deductions via a table schedule (cents in the dollar by price range). ³⁷

Shift detection and domain adaptation techniques that fit this setting

This project has **two distinct adaptation problems**, and each calls for different methods.

Shift type A: Betfair time drift (your features drift over time)

Even if you always use Betfair as the live feature source, the distribution of market states changes with seasonality, venue mix, liquidity regimes, and rule changes. Time-series domain adaptation work treats this as temporal covariate shift and proposes explicit temporal distribution matching. ³⁸

Practical detection signals:

- two-sample tests on rolling windows of core features (spread, depth, volume slope)
- drift detectors over streaming features (e.g., window-based independence tests, error-rate based detectors if you have fast labels) ³⁹
- calibration drift of your probability model (reliability degradation) using temperature scaling as a simple recalibration baseline ⁴⁰

Mitigation methods:

- **importance weighting / covariate-shift correction** when you can estimate density ratios $w(x) = p_{\text{target}}(x)/p_{\text{source}}(x)$. Covariate shift weighting is a classic approach. ⁴¹
A key caveat: deep nets can become insensitive to importance weights late in training; early stopping and regularisation choices matter. ⁴²
- **domain-invariant representations** (feature alignment) if you treat “time regime” as the domain label:
 - DANN-style adversarial invariance aligns feature representations that confuse a domain discriminator. ⁴³
 - CORAL-style alignment matches second-order statistics (covariances) across domains with low engineering overhead. ⁴⁴
- **time-series-specific adaptation:**
 - AdaRNN formulates temporal covariate shift explicitly and matches temporal distributions. ⁴⁵
 - DAF (Domain Adaptation Forecaster) shares attention components across domains while keeping some domain-private components. ⁴⁶
 - Recent test-time adaptation methods for forecasting (e.g., parameter-efficient TTA) aim to adapt online with unlabeled target streams. ⁴⁷

Shift type B: Betfair → TAB “price translation” drift (your unobserved target changes)

The mapping from Betfair state to TAB quote can drift due to promotions, price-setting heuristics, late liquidity, jurisdictional limits, or changes in tote pool dynamics. TAB rules allow price setting changes without notice and discretionary acceptance, so drift is structurally plausible. ¹³

Because you cannot observe TAB continuously, the practical approach is **selective observation**:

- maintain a small but representative panel of “labelled” events where you manually capture TAB quotes at specific times (e.g., T-15 min, T-5 min, at placement)
- use those labels to train and continually update the translation model
- treat this as a semi-supervised domain adaptation / missing-target-data problem where you have abundant unlabeled Betfair streams and scarce labelled TAB snapshots

Active learning provides a principled way to pick which events to label (e.g., high uncertainty in predicted TAB odds, or high expected value if your bet decision hinged on the translation). ⁴⁸

Label shift detection and correction

Label shift (change in class priors $p(y)$ between training and deployment with stable $p(x | y)$) is plausible if your historical training set over-represents certain race types or if your logging policy changes what “counts” as an example. BBSE (Black Box Shift Estimation) gives a way to detect and correct label shift using a classifier’s confusion matrix and predicted label marginals on target data. ⁴⁹

In this betting context, BBSE is most useful when:

- you frame y as a discretised label (e.g., favourite-bin, outcome category, or odds bucket), or
- you use calibrated probability outputs and can treat them as soft labels for marginal estimation tests.

Synthetic data and simulation (what it can and cannot do)

Simulation is useful for stress-testing policies under controlled “TAB slippage models” before you have dense TAB labels.

A pragmatic synthetic generator for TAB odds should encode:

- a deterministic component mapping Betfair implied probability to a bookmaker/tote implied probability
- a margin model (overround or take-out)
- a rounding/price-grid model
- a time-to-start component (late moves)
- a refusal/limit model as a censoring mechanism (TAB refuses or caps some bets). ⁵⁰

For higher-fidelity time-series augmentation, TimeGAN is a well-cited approach for generating realistic time series by combining adversarial and supervised sequence losses; reviews summarise broader GAN variants for time series. ⁵¹

Synthetic data should not be treated as validation; it is a **scenario generator**. Time-series UDA benchmarks emphasise that many methods behave differently once temporal dynamics are respected, so synthetic realism matters. ⁵²

Robust decision-making with uncertainty

Because TAB odds are partially unobserved, the system should produce **decision bounds** rather than point estimates:

- estimate a distribution for TAB odds given Betfair state (e.g., quantile regression)
- bet only if EV is positive at a conservative quantile (e.g., 10th percentile odds)
- track coverage and recalibrate bounds as labels arrive

Conformal prediction under covariate shift provides a way to build prediction sets/intervals with guarantees when you can estimate likelihood ratios using unlabeled target covariates. ⁵³

Practical TAB feature acquisition options and legal/ethical constraints

Official TAB pathways

TAB Studio presents a Web Services API that can retrieve up-to-date wagering information and place bets (single, batch, multi), and it has distinct pathways for account holders (personal) and commercial users. ²³

The TAB Web Services Terms of Use for account customers impose strong constraints: personal, non-commercial use only; restrictions on sublicensing, storage beyond temporary, derivative works, and excessive requests; and TAB retains discretion to refuse/suspend/terminate access. ⁵⁴

If your “betting app” is intended for any use beyond strictly personal, these terms strongly suggest you need a commercial agreement rather than relying on personal API access. ⁵⁵

Scraping and OCR (high-risk)

TAB’s general conditions explicitly state that “reproduction... including by screen scraping, is strictly prohibited” in relation to account channels and materials. ⁵⁶

TAB Web Services terms also limit copying, storage, and derivative works from “TAB Data,” even for API users. ⁵⁴

Given those explicit prohibitions, a strategy that automates extraction from TAB websites or app screens is legally and contractually risky, even if technically feasible. A safer alternative is to restrict any “capture” to **your own bet receipts and your own manually entered accepted odds**, focusing on data you are entitled to record for personal bookkeeping, and to obtain legal advice if you plan to operationalise beyond personal use.

Race fields and racing data licensing

Australian racing materials are heavily licensed. TAB’s web services terms list multiple copyright owners for racing materials (including Racing Information Services Australia ⁵⁷ and state racing bodies) and state that unauthorised reproduction/adaptation/communication is prohibited. ⁵⁴

Race-fields legislation and approvals frameworks exist; for instance, Harness Racing NSW ⁵⁸ explains that publishing NSW race fields without authorisation became an offence under race fields legislation. ⁵⁹

The Australian government's business licensing portal also frames "race field information use approval" as generally required for wagering operators, with conditions set by controlling bodies. ⁶⁰

If your product ever displays race fields or odds publicly, or if it is "providing a gambling service" rather than personal decision support, you should assume licensing/approvals will be required.

Gambling regulation signals that impact product design

Federal law (Interactive Gambling Act) prohibits some online services (e.g., online in-play sports betting) and advertising of prohibited services; enforcement sits with the Australian Communications and Media Authority ⁶¹, which publishes guidance and enforcement actions.

State regulators also state that online wagering is legal when providers are licensed in an Australian state/territory and reference the ACMA's enforcement role. ⁶²

If your app is purely for your own use and does not offer wagering services to others, the regulatory posture differs from that of an operator. Still, product decisions like "live in-play prompts" deserve caution in Australia because in-play online sports betting restrictions are explicit. ⁶³

Evaluation and validation strategy

What needs to be evaluated

You have three coupled performance targets:

1) **Probability quality** (does the outcome model estimate win probability well?) 2) **Translation quality** (does the TAB-odds model approximate the odds you can actually get?) 3) **Decision quality** (does the combined policy produce positive realised value net of frictions?)

Metrics

Probability model (classification / probabilistic forecasting):

- log loss, Brier score
- calibration metrics (ECE / reliability diagrams), with temperature scaling as baseline recalibration ⁴⁰

Translation model (odds regression / probabilistic regression):

- MAE/RMSE on log-odds or implied probability
- quantile loss (pinball loss) if you predict lower bounds
- coverage of prediction intervals if you use conformal bounds under shift ⁶⁴

Decision model (betting policy):

- ROI / yield (profit over stake), profit factor, maximum drawdown
- calibration of expected value vs realised value
- slippage analysis: distribution of "predicted TAB odds – accepted TAB odds"

- sensitivity to rule events (non-runners and deductions): reconcile your modelled adjustments with published tables/rules. ⁶⁵

Validation designs

Backtesting

Backtesting should replay Betfair streams at the same temporal resolution you use live and apply your decision rule at realistic decision times. Betfair historical stream data is available in time-stamped JSON tar files with multiple access tiers, which supports this replay style. ⁶⁶

If TAB data is sparse, backtests should incorporate uncertainty by sampling from your translation distribution and reporting performance envelopes rather than a single curve.

Shadow-mode live testing

Run the model live, generate recommendations, but do not bet (or bet minimal stakes) while you measure:

- drift in Betfair feature distributions
- stability of your probability outputs
- difference between predicted TAB odds and manually observed TAB odds (on a sampled subset). ⁶⁷

A/B testing

A/B is plausible only if you can randomise between strategies under similar conditions and log accepted odds and outcomes reliably. In manual execution settings, A/B often becomes “alternating days/races” or “randomised candidate selection sets” to reduce operator bias.

Monitoring for concept drift

In streaming environments, drift detectors and window-based tests are standard; methods like adaptive windows and distribution tests are well established, with newer work emphasising non-parametric approaches. ⁶⁸

Operationally, drift monitoring is easiest on features you always observe (Betfair), and hardest on those you rarely observe (TAB). A practical compromise:

- continuous drift monitoring on Betfair features
- periodic “audit labels” for TAB odds translation using active learning sampling ²⁶
- alerting on business KPIs (hit rate vs expected, EV vs realised, slippage spikes)

Open-source drift tooling (e.g., SeldonIO ⁶⁹’s alibi-detect) provides detectors for tabular and time-series drift. ⁷⁰

Implementation steps, tooling, and timeline

Concrete implementation steps

Step zero: define the decision contract

Lock down what a recommendation means operationally:

- product type: fixed odds vs tote
- when the user will place the bet (T-X minutes, at jump, etc.)

- what the user must record (accepted odds, time, stake, any refusal/reprice)

This is driven by the fact that fixed odds and tote are fundamentally different instruments. ⁷¹

Build the Betfair ingestion spine

Use streaming to maintain an in-memory market state and persist snapshots at decision-relevant times. Betfair describes the stream API as low-latency and listMarketBook/stream as providing ladders, traded ladders, LTP, and market-def updates; virtual prices may lag non-virtual. ¹

For development, Betfair offers delayed keys; live access requires approval and a one-off fee for a live key.

⁷²

Baseline probability model

Start with a model that predicts win probability from:

- Betfair-derived prices/volumes/stability features
- stable covariates (event-type, track class, runner metadata)

Treat this as a time-series learning problem if you use trajectories (e.g., sequences of price/volume changes), and consider time-series adaptation methods if you will retrain across regimes. ⁷³

Translation model v1 (manual labels)

Create a minimal “TAB odds capture” workflow:

- select N races per week for full capture of all runners’ fixed odds and/or tote approximates at pre-defined timestamps
- record the accepted odds at the moment of bet placement
- store refusal/reprice events

TAB’s own documentation clarifies that tote approximates and prize pools are part of the published information set and that tote fluctuations are shown shortly before advertised start time; these cues can inform your capture schedule. ⁷⁴

Active learning loop

Implement acquisition rules like:

- “label if translation uncertainty is high”
- “label if model is about to recommend a bet near the decision threshold”
- “label if drift monitors flag a regime change”

Active learning surveys provide standard query strategies for uncertainty-based selection. ⁴⁸

Robust decision rule

Move from point estimates to conservative rules:

- convert predicted win probability and predicted TAB-odds distribution into a conservative EV (e.g., lower quantile odds)
- apply bankroll constraints and max exposure per event

- explicitly incorporate rule adjustments: Betfair reduction factors and TAB deductions schedules differ and must be applied correctly in evaluation and settlement logic. ⁷⁵

Monitoring and retraining

Add:

- feature drift monitors (Betfair)
- calibration monitors (probability model) with temperature scaling as a quick recalibration tool ⁷⁶
- translation residual monitors on labelled TAB snapshots
- periodic retraining and/or test-time adaptation for non-stationary regimes (parameter-efficient approaches aim to adapt with limited updates). ⁷⁷

Recommended open-source tools and libraries

Betfair integration:

- `betfairlightweight` (Python wrapper with streaming and betting operations) ⁷⁸
- `flumine` (event-based sports betting exchange trading framework; stream handling and execution patterns) ⁷⁹
- Betfair official API-NG sample code repository ⁸⁰

Domain adaptation / transfer tooling:

- `PyTorch Adapt` for domain adaptation pipelines and adapters ⁸¹
- `ADAPT` (Python toolbox for transfer learning and domain adaptation) ⁸²
- THUML Transfer-Learning-Library (broad DA/finetune setups) ⁸³

Drift and monitoring:

- `alibi-detect` for drift detection on tabular/time series ⁸⁴

Synthetic time series:

- TimeGAN reference implementation repository (for controlled experiments only) ⁸⁵

Timeline (mermaid Gantt)

```
gantt
    dateFormat YYYY-MM-DD
    title Betfair→TAB domain adaptation delivery plan

    section Foundations
        Define decision contract & logging UX           :a1, 2026-02-18, 7d
        Betfair streaming ingestion + storage           :a2, after a1, 14d

    section Baselines
        Baseline outcome model (probabilities)         :b1, after a2, 21d
        Baseline monitoring (feature + calibration)     :b2, after b1, 14d
```

section TAB translation via human labels	
Manual TAB capture workflow v1	:c1, after a1, 14d
Translation model v1 (Betfair→TAB odds)	:c2, after c1, 21d
Active learning sampling rules	:c3, after c2, 14d
section Robust decisioning	
Conservative EV + uncertainty bounds	:d1, after c2, 21d
Backtest harness + scenario stress tests	:d2, after d1, 21d
section Commercialisation path	
Apply for commercial TAB data access	:e1, 2026-02-18, 56d
Vendor evaluation / contracting (if needed)	:e2, after e1, 42d

Key papers to read (APA-style, minimal list)

- Yaroslav Ganin ⁸⁶ et al. (2016). *Domain-Adversarial Training of Neural Networks*. Journal of Machine Learning Research. ⁸⁷
- Baochen Sun ⁸⁸ & Kate Saenko ⁸⁹ (2016). *Deep CORAL: Correlation Alignment for Deep Domain Adaptation*. ⁹⁰
- Zachary C. Lipton ⁹¹ et al. (2018). *Detecting and Correcting for Label Shift with Black Box Predictors*. ICML. ⁹²
- Hidetoshi Shimodaira ⁹³ (2000). *Improving predictive inference under covariate shift by weighting the log-likelihood function*. Journal of Statistical Planning and Inference. ⁹⁴
- Masashi Sugiyama ⁹⁵ & Motoaki Kawanabe ⁹⁶ (2012). *Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation*. ⁹⁷
- Ryan J. Tibshirani ⁹⁸ et al. (2019). *Conformal Prediction Under Covariate Shift*. ⁹⁹
- Yue Du ¹⁰⁰ et al. (2021). *AdaRNN: Adaptive Learning and Forecasting of Time Series*. CIKM. ¹⁰¹
- Xiang Jin ¹⁰² et al. (2021). *Domain Adaptation for Time Series Forecasting via Attention Sharing*. ¹⁰³
- Burr Settles ¹⁰⁴ (2009). *Active Learning Literature Survey*. ¹⁰⁵

Primary-source operational documents to keep near your codebase

- Betfair developer docs on exchange APIs (Betting API, Accounts API, Stream API; low-latency stream positioning). ¹⁷
- Betfair developer support on what market data is available (ladders, traded ladders, LTP, SP projections; virtual price update latency). ¹⁹
- Betfair's published odds increment table (tick ladder). ¹⁴
- TAB documents on tote mechanics (pool, take-out/commission, approximates). ³⁶
- TAB fixed odds description and fixed-odds rules/discretion language. ¹⁰
- TAB fixed-odds deductions schedule for scratchings (NSW/VIC). ²¹
- TAB Web Services Terms of Use and Conditions of Use (personal vs commercial use constraints; screen scraping prohibition). ¹⁰⁶
- Australian Communications and Media Authority ⁶ guidance on Interactive Gambling Act restrictions if your product edges toward "service provision" rather than personal tooling. ⁶³

1 17 57 **Guide to the Betfair Exchange API | Betfair Developers**

<https://developer.betfair.com/exchange-api/>

2 9 31 36 58 71 **TAB Tote Win and Place Bet - Help Centre**

https://help.tab.com.au/betting-on-racing/tab-tote-win-and-place-bet?utm_source=chatgpt.com

3 **Domain Adaptation for Time Series Forecasting via ...**

https://arxiv.org/abs/2102.06828?utm_source=chatgpt.com

4 23 24 69 86 88 89 **Tab Studio**

<https://www.studio.tab.com.au/features>

5 53 64 **[1904.06019] Conformal Prediction Under Covariate Shift**

https://arxiv.org/abs/1904.06019?utm_source=chatgpt.com

6 12 **What Approximate Dividend Information is Available**

<https://help.tab.com.au/betting-on-racing/what-approximate-dividend-information-is-available>

7 61 63 **About the Interactive Gambling Act**

https://www.acma.gov.au/about-interactive-gambling-act?utm_source=chatgpt.com

8 **Betfair How-To**

https://betfair-datascientists.github.io/wagering/betfairHowTo/?utm_source=chatgpt.com

10 37 **TAB Fixed Odds Win and Place Bet - Help Centre**

https://help.tab.com.au/betting-on-racing/tab-fixed-odds-win-and-place-bet?utm_source=chatgpt.com

11 19 28 32 33 34 35 98 **What Betfair Exchange market data is available from listMarketBook & Stream API? – Betfair Developer Program**

<https://support.developer.betfair.com/hc/en-us/articles/6540502258077-What-Betfair-Exchange-market-data-is-available-from-listMarketBook-Stream-API>

13 29 30 50 102 **TAB LIMITED FIXED PRICE RACING BETTING RULES**

https://www.tabinfo.com.au/documents/NSW_TAB%20Limited%20Fixed%20Price%20Rules-FINAL%2009%20September%202017.pdf?utm_source=chatgpt.com

14 95 **Betting on the Internet**

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